Extensions: mixed responses, spatial and temporal ordination

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Outline

- Mixed response types: modelling communities with different measurement scales
- Spatial and temporal autocorrelation in latent variables
- Case study: joint modelling across survey methods with spatial and temporal structure

Questions so far?

Not all species are measured the same way

Community data rarely comes in a single format:

- Species composition: presence-absence
- Dominant grasses: % cover
- Rare forbs: count of individuals
- Invertebrates: biomass

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Standard GLLVMs fit one distribution to all species. But forcing binary, count, and continuous responses into the same family means either:

- **Information loss** (discretising continuous data)
- **Model misspecification** (Poisson for proportion data)

Mixed response GLLVMs

In `gllvm`, the `family` argument accepts a **character vector** of length p (one per species column):

```
family_vec <- c(rep("binomial", 12), rep("poisson", 8), rep("

fit <- gllvm(y = Y, family = family_vec, num.lv = 2)
```

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```

Each species gets its own link function and variance structure. The latent variables $\mathbf{u}_i$ are shared across all species.

What distributions are available?

Family	Data type	Link
`"gaussian"`	Continuous, symmetric	identity
`"tweedie"`	Non-negative, zero-inflated	log
`"orderedBeta"`	Proportions including 0/1	logit
`"betaH"`	Proportions excluding 0 or 1	logit
`"poisson"`	Counts	log
`"negative.binomial"`	Overdispersed counts	log
`"binomial"`	Presence-absence or trials	probit
`"ZIP"`, `"ZINB"`	Zero-inflated counts	log

Why the latent variable framework makes this possible

The GLLVM is species-specific in its response model:

$$g_j(\mu_{ij}) = \beta_{0j} + \mathbf{x}_i^\top \boldsymbol{\beta}_j + \mathbf{u}_i^\top \boldsymbol{\gamma}_j$$

but the latent variables $\mathbf{u}_i$ are **shared**: they represent the underlying community state at site i.

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Different families for different j is just a change in $g_j(\cdot)$ and the variance function; the ordination structure is unaffected.

This is what makes joint modelling across survey methods possible.

Binomial with $N > 1$ trials

When species are recorded as frequencies (e.g., Raunkiaer: k quadrats occupied out of $N = 100$):

```
fit <- gllvm(y = Y, family = "binomial", Ntrials = 100, num.l
```

`Ntrials` can also be a vector of length p if trials differ per species, or a matrix of size $n \times p$.

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Default link for binomial in `gllvm` is **probit**, not logit.

orderedBeta for proportions

Proportions including exact 0 and 1 (e.g., Braun-Blanquet cover converted to $[0,1]$):

```
fit <- gllvm(y = Y / 100, family = "orderedBeta", num.lv = 2)
```

The orderedBeta distribution places point masses at 0 and 1 with a Beta distribution on $(0,1)$. More appropriate than `"beta"` when true absences and full cover both occur.

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`zetacutoff` controls the quantile threshold for the zero/one mixture; default is usually fine, but set explicitly for stability:

```
fit <- gllvm(..., family = "orderedBeta", zetacutoff = c(-2,
```

Combining families in a single model

A realistic community dataset might need:

```
family_vec <- ifelse(grepl("_cover$", colnames(Y)), "orderedB
              ifelse(grepl("_count$", colnames(Y)), "negative
                                                    "binomial

Ntrials <- ifelse(grepl("_freq$", colnames(Y)), 100, 1)

fit <- gllvm(y       = Y,
             family = family_vec,
             Ntrials = Ntrials,
             num.lv = 2,
             zetacutoff = c(-2, 2))
```

What does a joint model across datasets look like?

Two surveys, different methods, stacked as rows:

- Rows 1–n_1: survey A (ordinal columns only)
- Rows $n_1 + 1$–$n_1 + n_2$: survey B (binomial columns only)
- Shared species appear twice: `_COV` and `_FREQ` columns, with `NA` in the opposite block

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The model estimates **one set of latent variables** covering all rows, and **separate loadings** for `_COV` and `_FREQ` versions of each shared species.

If the methods agree, $\hat{\gamma}_{j,\mathrm{COV}} \approx \hat{\gamma}_{j,\mathrm{FREQ}}$; testable with cosine similarity or Procrustes.

A core assumption we have not questioned

All GLLVMs so far assume sites are **independent** after conditioning on the latent variables and covariates.

$$\mathbf{u}_i \overset{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

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This fails when:

- Plots are close in space and share unmeasured environmental variation
- The same plots are revisited over time
- The design has nested structure (transects within sites)

Structured covariance for the latent variables

Replace independent LV draws with a **correlated** distribution:

$$\mathbf{u}_{.q} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_q), \quad q = 1, \dots, d$$

where $\mathbf{\Sigma}_q$ encodes the dependence structure.

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In `gllvm`: argument `lvCor` specifies the covariance model; `studyDesign` provides the grouping or ordering variable.

Structures available in gllvm

Formula syntax	Structure	Use case
`~corAR1(1|time)`	AR(1)	Equally-spaced time series
`~corExp(1|site)`	Exponential kernel	Spatial data (coordinate-based)
`~corMatern(1|site)`	Matérn kernel	Spatial data (smoother fields)
`~corCS(1|group)`	Compound symmetry	Nested designs

The same structures are available for row effects via `row.eff`.

You have seen `lvCor` before for nested designs (`corCS`); the same argument accepts spatial and temporal kernels.

Temporal autocorrelation: AR(1)

Community composition at time t is correlated with time $t-1$:

$$\boldsymbol{\Sigma}_q[t, t'] = \rho^{|t-t'|}, \quad \rho \in (-1, 1)$$

```
fit_ar1 <- gllvm(
  y           = Y,
  X           = Xenv,
  formula     = ~ temperature + precipitation,
  num.lv      = 2,
  studyDesign = data.frame(site = site_id, year = year_id),
  row.eff     = ~corAR1(1|year),
  lvCor       = ~corAR1(1|year),
  family      = "negative.binomial"
)
```

Spatial autocorrelation: corExp

Correlation decays exponentially with geographic distance:

$$\mathbf{\Sigma}_q[i,k] = \exp\left(-\frac{d_{ik}}{\rho}\right)$$

ρ is the **range parameter**: the distance at which correlation has decayed to $e^{-1} \approx 0.37$.

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Effective range (correlation < 0.05): $d \approx 3\rho$.

```
fit_sp <- gllvm(
  y            = Y,
  num.lv       = 2,
  distLV       = site_coords,    # n x 2 coordinate matrix
  lvCor        = ~corExp(1|site),
  studyDesign = data.frame(site = site_id),
```

Spatial vs. temporal: which to use?

Row effects (`row.eff`)
Community-level variation: affects all species equally at a site.
Good for: site productivity, observer effects, year-level abundance shifts.

LV correlation (`lvCor`)
Species composition variation: different species respond differently to the latent gradient.
Good for: spatially or temporally structured community turnover.

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Species composition variation: different species respond differently to the latent gradient.
Good for: spatially or temporally structured community turnover.

Both can be included simultaneously:

```
fit <- gllvm(...,
  row.eff = ~corAR1(1|year),
  lvCor   = ~corExp(1|site))
```

The Matérn covariance

A more flexible spatial kernel:

$$K(d) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}\, d}{\rho} \right)^{\nu} \mathcal{K}_\nu \left(\frac{\sqrt{2\nu}\, d}{\rho} \right)$$

- $\nu = 1/2$ recovers the exponential kernel
- Higher ν produces smoother spatial fields
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Use `corMatern` when you expect smooth spatial gradients rather than abrupt local variation.

Computational considerations

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`gllvm` uses **sparse approximations** to the variational covariance matrix:

- `Lambda.struc = "bdNN"`: block-diagonal nearest-neighbour Cholesky
- `Lambda.struc = "UNN"`: Kronecker product with NN sparse Cholesky

`NN` controls the number of neighbours; more neighbours = better approximation, slower fitting.

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For large datasets ($n > 500$), start with `NN = 5` and increase if needed.

Concurrent ordination with covariates

Yesterday we saw **concurrent ordination**: LVs constrained to be linear combinations of predictors.

With `num.lv.c` and `lv.formula`, we can combine this with spatial/temporal structure:

```
fit_conc <- gllvm(
  y           = Y,
  X           = X,
  family      = "poisson",
  num.lv.c    = 2,
  lv.formula  = ~ temperature + moisture,
  distLV      = site_coords,
  lvCor       = ~corExp(1|site),
  studyDesign = data.frame(site = site_id),
```

Case study, Garchinger Heide

```
## List of 3
##  $ coverage :List of 3
##   ..$ Y    : num [1:84, 1:80] 0 0 0 0 0 0 0 0 0 0 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   ..$ Y_ord: int [1:84, 1:80] 1 1 1 1 1 1 1 1 1 1 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   ..$ X    :'data.frame':    84 obs. of  3 variables:
##  $ frequency:List of 2
##   ..$ Y: num [1:120, 1:116] 0 0 0 0 0 0 0 0 0 0 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   ..$ X:'data.frame':    120 obs. of  3 variables:
##  $ coords   :List of 2
##   ..$ coverage :'data.frame':    42 obs. of  3 variables:
##   ..$ frequency:'data.frame':    40 obs. of  3 variables:
```

Families used in the Garchinger case study

Family	Data type	Link
`"ordinal"`	Ordered categories (Londo / BB codes)	cumulative probit
`"ZNIB"`	Zero-inflated count, N trials	log

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`zeta.struc = "common"`: cutpoints shared across species. Species-specific cutpoints (`"species"`) are unidentifiable under a block-diagonal NA structure.

The Garchinger Heide dataset

A calcareous grassland in Bavaria (Bauer & Albrecht 2020, Basic and Applied Ecology; data: PANGAEA).

Two long-term monitoring programmes, never combined before:

Coverage survey
42 plots $\times$ $\{2003, 2018\}$
Londo ordinal codes (12 levels)
$\rightarrow$ ordinal

Frequency survey
40 plots $\times$ $\{1984, 1993, 2018\}$
Raunkiaer (/100 quadrats)
$\rightarrow$ ZNIB, $N = 100$

Question: do the two methods recover the same community gradient?

The joint model design

Stack both surveys as rows; shared species get two columns:

	Bromus_COV	Bromus_FREQ	Molinia_FREQ
COV, plot S03, 2003	5	NA	NA
COV, plot S03, 2018	7	NA	NA
FREQ, plot II5, 1984	NA	73	12
FREQ, plot II5, 2018	NA	81	8

- 204 rows, 196 columns; 79 shared species $\times$ 2 + survey-only
- Block-diagonal NA structure: no plot is in both surveys

The identification problem

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But rotation ambiguity means we cannot guarantee the two blocks are in the same orientation.

We need an external anchor.

Model 1: shared species only

```
fit3 <- gllvm(
  y           = Y_joint,
  family      = family_vec,
  Ntrials     = Ntrials,
  num.lv      = 2,
  zeta.struc  = "common",
  row.eff     = ~(1|site),
  studyDesign = data.frame(site = physical_site)
)
```

Diagnose alignment: cosine similarity between $\hat{\gamma}_{j,\mathsf{COV}}$ and $\hat{\gamma}_{j,\mathsf{FREQ}}$ for each shared species.

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A bimodal distribution at ± 1 signals rotation ambiguity. Non-bimodal signals real disagreement.

Model 2: spatial anchor

4 coverage–frequency plot pairs lie within 6 m of each other. The spatial covariance provides a partial anchor via nearby plots.

```
fit4 <- update(fit3,
  distLV        = coords_sites,
  lvCor         = ~corExp(1|site),
  studyDesign   = data.frame(site = physical_site),
  Lambda.struc = "UNN")
```

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```

`Lambda.struc = "UNN"`: sparse nearest-neighbour Cholesky approximation; see computational considerations slide.

Results: spatial model

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- Correlation ≈ 0.2, protest $p = 0.081$
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Fitted spatial range: $\hat{\rho}_{\text{LV1}} \approx 15$ m, $\hat{\rho}_{\text{LV2}} \approx 2481$ m.

Effective range (Cor < 0.05): ~ 44 m and ~ 7432 m.

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Effective range (Cor < 0.05): ~ 44 m and ~ 7432 m.

Most inter-plot distances exceed 80 m; the community gradient is driven by local habitat conditions, not broad spatial autocorrelation. The few close pairs provide just enough anchor.

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Disagreement is concentrated in species that have **changed between the survey periods** (1984–1993 vs. 2003–2018), not random noise.

This is a resurvey: disagreement $=$ ecological change, not methodological failure.

Model 3: adding temporal structure

Year as a concurrent covariate; site random intercepts carry over from fit4:

```
fit5 <- update(fit4,
  num.lv         = 0,
  X              = X_joint,
  num.lv.c       = 2,
  lv.formula     = ~ Year,
  randomB        = "LV",
  lvCor          = ~corExp(1|site),
  Lambda.struc = "UNN")
```

What does each component do?

- `lv.formula = ~ Year`: LVs are driven by survey year; temporal community change is captured in the ordination axes
- `row.eff = ~(1|site)`: site-level random intercepts absorb overall abundance differences across plots
- `lvCor = ~corExp(1|site)`: residual spatial structure in community composition after accounting for Year

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- `lvCor = ~corExp(1|site)`: residual spatial structure in community composition after accounting for Year

The residual loadings alignment then answers: after partialling out temporal change, do the two surveys still agree on the community gradient?

Three model perspectives

Model	Anchor	Alignment	Interpretation
fit3: shared species	Shared γ only	Bimodal ± 1?	Rotation ambiguity
fit4: + Spatial	Nearby cross-survey pairs	$r \approx 0.2$	Methods agree spatially
fit5: + Year	Space + time	$r \approx 0.04$	Residual agreement

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The three models are not competitors: they answer **different questions** about the same data.

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The joint family GLLVM makes all three possible within a single framework.

Model comparison

```r
AIC(fit3, fit4, fit5)
```

```
##       df      AIC
## fit3 830 65243.53
## fit4 832 72023.04
## fit5 834 66567.71
```

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```

AIC favours the more complex models, but the models target different questions: do not use AIC to choose among them.

Summary

Mixed responses

- `family` accepts a vector; each species gets its own distribution and link
- Shared LVs connect all species into one ordination space
- Enables joint modelling across datasets with different measurement scales

Spatial and temporal autocorrelation

- `lvCor` and `row.eff` accept structured covariance models (`corExp`, `corAR1`, ...)
- `distLV` provides coordinates for spatial kernels
- Short fitted ranges are informative, not a failure

Combining both

- Mixed responses + spatial LVs + concurrent ordination: all composable in `gllvm`
- The Garchinger example shows how far this can be pushed on real data